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Measuring the Strength of Attitudes in Social Media Data

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5.1 Introduction

The amount of stored data is expected to double every two years for the next decade (Quartz 2015). The advances in social science research will be enormous even if only a fraction of these data can be used. Official statistics could be produced more quickly and less expensively than currently possible using surveys (Kitchin 2015). Big Data may also provide statistics for small areas to make informed policy and program decisions (Marchetti et al. 2015). Researchers have already begun to harness Big Data from social media for a variety of purposes, including research into disease prevalence, consumer confidence, and household characteristics (Butler 2013; O'Connor et al. 2010; Citro 2014).

In researchers' excitement to capitalize on Big Data, many have published papers in which they attempt to create point estimates from Big Data. Unfortunately, there are as many examples of biased estimates created from Big Data as there are success stories. For example, Twitter data has successfully been used to predict the 2010 UK, 2011 Italian, and 2012 French elections (Kalampokis et al. 2017; Ceron et al. 2014). However, Chung and Mustafaraj (2011) found Twitter to be an unreliable source to predict the 2010 US Senate special elections in Massachusetts while Gayo-Avello (2013) was unable to reproduce Ceron and colleagues' findings using similar data and methods. Similarly, in 2008, Google Flu Trends was touted as being able to successfully predict the number of individuals suffering from the flu (Ginsberg et al. 2009). However, in 2009 and again in 2013, this same data source severely misestimated flu prevalence (Butler 2013; Cook et al. 2011). In a final example, researchers have used Twitter

Big Data Meets Survey Science: A Collection of Innovative Methods, First Edition. Edited by Craig A. Hill, Paul P. Biemer, Trent D. Buskirk, Lilli Japec, Antje Kirchner, Stas Kolenikov, and Lars E. Lyberg.
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data to successfully measure and track change in consumer sentiment over time (O'Connor et al. 2010; Bollen, Mao, and Pepe 2011). However, these trends broke down after 2011 (Pasek et al. 2018; Conrad et al. 2020).

Each example of creating point estimates using Big Data, regardless of success, is useful in helping researchers think outside the box and find more creative ways to collect and analyze data. However, these types of papers are not generalizable. They use one type of Big Data to produce one estimate using one type of model. When Big Data can be manipulated to produce unbiased estimates, consideration is rarely given to why it worked. When the application fails to produce unbiased estimates, most researchers have limited their explanation of the failure to untestable hypotheses or specific reasons that are, once again, not generalizable to other applications.

It is critical to understand the underlying reasons for the successes and failures of different applications of Big Data to allow researchers more foresight into the ways in which Big Data may be used and the ways it may not (Japec et al. 2015). To this end, several researchers have developed theoretical frameworks that decompose error from Big Data sources into components. Frequently cited error components include (i) missingness when individuals do not subscribe to the given data source; (ii) inappropriate use of the data (i.e. the data were not intended for the researcher's purpose so may measure the researcher's preferred construct); and (iii) researcher's inaccurate modeling or coding (Baker 2017; Schober et al. 2016; Biemer 2016; Hsieh and Murphy 2017). However, these frameworks are mostly theoretical and have not been applied to actual applications to identify how much each of these errors is contributing to the overall error.

Another limitation of the existing literature is its focus on proportions, prevalence estimates, and other dichotomous outcomes (e.g. election outcomes and flu prevalence).[1] These data points are important in official statistics. However, researchers often depend on other types of data, such as scales used, to measure attitude strength. For example, dichotomous indicators are useful to predict the outcome of an election, but scales are useful to identify swing votes during the campaign. Similarly, sociologists depend on scales to identify the types of individuals who may be affected by a social intervention. Individuals who fall on the extreme ends of the spectrum are unlikely to be swayed by politicians or affected by a social intervention, whereas individuals who feel less strongly may be more likely to change their opinion (Dalege et al. 2017; Bos, Sheets, and Boomgaarden 2018; Doherty and Adler 2014). Therefore, it is important to accurately measure attitude strength and understand the potential for measurement error on such scales.

In this chapter, we extend the current literature by attempting to use social media data to produce similar attitude distributions as survey data. We then dive

1 An exception to this statement is the consumer sentiment analysis described above.

deeper into each distribution to determine the source of any observed dissimilarities. We use a combination of several datasets stemming from the European Social Survey (ESS) and Reddit to measure six scalar attitudes (political ideology, interest in politics, support for gay rights, support for the European Union [EU], views on immigration, and belief in climate change). Error was evaluated both overall, and by three commonly cited error components of Big Data: (i) missingness, (ii) measurement, and (iii) coding.

5.2 Methods

5.2.1 Data

To address our first objective, producing attitude distributions using social media data, we compared sentiment scores derived from Reddit comments and posts made between October 2016 and March 2017 to survey estimates from the 2016 German ESS. Reddit.com is the self-proclaimed "front page of the Internet." Registered users may post text, a link, images, or video onto any of thousands of subreddits (individual pages that cover a given topic, originally designed to be similar to a section of a newspaper). Other individuals may read, vote them up or down, or comment on the post. Although anyone may read Reddit content (to be sure, Reddit receives 14 billion screen views per month, worldwide [https://www.redditinc.com]), one must login with an anonymous username to participate (e.g. submit a post or comment).

The ESS is a repeating cross-sectional survey that measures attitudes on a host of topics in several countries throughout Europe. Six sentiments or attitudes were compared: political ideology, interest in politics, gay rights, the EU, immigration, and climate change. Unfortunately, the attitude distributions produced using the Reddit posts and comments were not comparable to those observed in the ESS data (details to follow), and we turned our attention to the second objective – understanding the cause of the difference.

To address the latter objective, we isolated error due to missingness, measurement error, and coding error. In this chapter, we define these terms broadly. Missingness refers to any individual or attitude missing from the dataset. This conflates two types of errors from the Total Survey Error (TSE) framework, undercoverage, and nonresponse, and we attempt to separate these two types of missingness in the analysis (Biemer and Amaya 2020; Groves and Lyberg 2010; Groves 2004). Measurement error includes any data that do not accurately represent the desired construct. Again, this conflates specification error and traditional measurement error from TSE. Unfortunately, we cannot parse these error components in this analysis. Finally, coding error includes bias introduced by researcher mishandling of the data such as including off-topic posts or assigning the incorrect sentiment score to the Reddit text data. This definition is similar to the TSE definition, although it is less common in survey research since much of the "coding" is done by the respondent (e.g. respondents provides their own sentiment score on a predefined scale).

To isolate the different error components, we conducted a series of comparisons using data from two additional data sources: (i) a 2018 nonprobability survey conducting using the Reddit platform, and (ii) one year of Reddit submissions made by 2018 Reddit Survey respondents. Throughout this chapter, we refer to the various datasets as the ESS, Reddit 2016, Reddit Survey, and Reddit 2018, respectively. In the rest of this section, we describe each data source and how each was used in the analysis.

5.2.1.1 European Social Survey Data

In Germany, the ESS was conducted using a stratified, two-stage probability sample of 9456 residents 15 years or older drawn from the local residents' registration offices' official registers. A total of 2852 German in-person interviews were conducted between 23 August 2016 and 26 March 2017 using the standard ESS questionnaire (AAPOR RR1 = 30.61%; European Social Survey 2016). In this chapter, we limited our analysis to interviews conducted with individuals 16 years and older ($n = 2829$) to be comparable to Reddit users (who are restricted to 16 and older in Germany [https://www.redditinc.com/policies/user-agreement]) and maximize comparability.

Although the ESS covers a host of topics, we limited our analysis to data from questions covering six areas: political ideology, interest in politics, gay rights, the EU, immigration, and climate change. These topics were chosen to span a broad array of issues and were hypothesized to vary in our ability to identify Reddit submissions on the same topics. For example, people use a set of finite words to discuss climate change (e.g. climate change or global warming), whereas submissions to express the lack of political interest may not use a fixed set of words. Appendix 5.A includes a list of all questions used in analysis and the topic to which they belong. All questions were combined to create a singular index for each topic. Although all questions used in the analysis were asked on a scale, the scales varied (i.e. 2-point, 4-point, 5-point, or 11-point). Where necessary, factors were assigned to convert scales to an 11-point scale. Then, questions on a given topic were summed and divided by the count of items, creating an average score across items that ranged from 0 to 10. Mathematically, this took the form:

$$x_{q,\text{Revised}} = x_{q,\text{Original}} \times \frac{10}{s_q - 1}$$

$$X = \frac{\sum_1^Q x_{q,\text{Revised}}}{Q}$$

where $x_{q,\text{Original}}$ is the original sentiment value and s_q is the number of points on the scale for the question. Q is the total number of questions, and X is the overall sentiment score.

The ESS data were poststratified to population targets from the EU Labour Force Survey 2016 (Eurostat). Unless otherwise noted, the final weights that account for sample selection and poststratification were used in the analysis.

5.2.1.2 Reddit 2016 Data

We downloaded[2] all submissions made between October 2016 and March 2017 that originated from a selection of 367 subreddits, overlapping with approximately 85% of the ESS field period in Germany. The chosen subreddits were identified from a list that is meant to include all subreddits originating from German-speaking countries (i.e. Germany, Austria, Lichtenstein, Luxembourg, and Switzerland; DACH Reddit wiki page [https://www.reddit.com/r/dach/wiki/k]). This list is maintained by users, and there was no method to confirm its completeness. Moreover, our goal was to use Reddit data to make inferences about attitudes in Germany. However, given the anonymity of redditors, location was unknown. It is possible that individuals who participated on the selected subreddits do not live in Germany or that German residents posted on other subreddits. To the extent that the mismatch between the target universe of all German residents' submissions and our frame of all posts on the identified subreddits is systematic and not random, this method introduced bias into our analysis, which we discuss later.

The 367 included subreddits originated in German-speaking countries, but submissions were in multiple languages. We used R package CLD3 to identify the language of the submission: 68.7% were categorized as German, 10.6% as English, and 20.7% as undeterminable or language other than English or German. Only submissions categorized as German or English ($n = 463\,234$ submissions by 25 742 authors) were available for analysis. Including all posts would have improved coverage, but we excluded these because we assumed that these posts were unlikely to concern the six topics of interest.

We removed all punctuation and converted all text to lowercase to reduce error caused by case-sensitive searches. Each original post and its accompanying comments were collapsed into a single entry (as opposed to one entry for a post and one entry for each comment) for topic identification. A dictionary-based search was applied to the collapsed data to identify submissions on the six topics of interest (i.e. political ideology, interest in politics, gay rights, the EU, immigration, and climate change; Appendix 5.B). Comments and posts were initially collapsed to

2 Data were downloaded from http://files.pushshift.io/reddit. Reddit content has been scraped and posted monthly to this site since 2005.

ensure relevant material was not missed, given the brevity of some posts and comments and the widespread use of pronouns.[3] We then brainstormed search terms; we used a thesaurus to add more words and also reviewed additional words that Google recommended once a topic was entered into the search box. Some Boolean operators (e.g. "and" or "or") were used to minimize false positive and false negative errors.

All relevant submissions on a given topic were re-separated (one entry per post and one entry per comment), then collapsed by language and author, creating one entry per author per language per topic. The entries were tokenized, and a sentiment analysis was conducted using the AFINN Sentiment Lexicon for English entries (Nielsen 2011) and the SentimentWortschatz (SentiWS) Lexicon for German entries (Goldhahn, Eckart, and Quasthoff 2012). These lexicons were chosen because they provide weights for each word (as opposed to just labeling a word as positive or negative), which was critical to reproduce scaled attitude scores. They are also widely used lexicons in their respective languages. The team calculated sentiment for each individual and each topic as the average score of all words used by the author that were found in the lexicons. Scores could range from −5 to 5 and were recentered to range from 0 to 10 to be directly comparable to the ESS scale. Table 5.1 displays the counts of records at each stage of the coding process. All analyses were conducted using the cases in the final column – one record per author for which a sentiment score was calculated on a given topic. Sentiment

Table 5.1 Number of entries by type and summarization metric.

	Authors with data on relevant topics			Authors with measurable sentiment		
	German	English	Total	German	English	Total
Total	14 549	4 733	16 643	11 850	3 779	14 051
Political ideology	11 172	3 497	12 712	9 190	2 842	10 830
Interest in politics	13 416	4 181	15 217	10 929	3 312	12 826
Immigration	9 404	2 605	10 505	7 770	2 093	8 857
EU	5 500	1 204	6 036	4 489	958	5 005
Gay rights	7 777	1 701	8 452	6 453	1 307	7 107
Climate change	4 428	1 132	4 964	3 700	939	4 250

Source: Reddit (2016).

3 For example, an original post could be "I support Merkel," and a response could be "I support her, too." If we had run the dictionary-based search on each submission individually, we would have picked up the original post but not the comment. We collapsed them to determine if any mention was relevant, flagged all submissions within relevant posts than reformatted the data back to the one row per submission for later analysis (see Appendix 5.C).

scores for a given topic were possible if an author had at least one word that was found in either sentiment lexicon.

5.2.1.3 Reddit Survey

Between 29 June and 13 July 2018, we conducted a survey by posting a link on 244 of the 367 previously identified subreddits. One hundred twenty-three (123) subreddits were excluded because they were private, meant to teach German (i.e. the readers likely did not reside in a German-speaking country), did not allow posts that contained links, or the subreddit moderator explicitly asked us to exclude their subreddit from the survey. The survey included all questions used in the analysis of the ESS data with modification made as necessary to adapt for the difference in mode. To make the survey of interest to a wider population, we also included questions about Reddit at the beginning of the survey. Finally, we asked respondents to provide their Reddit username so it could be linked to their Reddit submissions. Although anyone who saw the link could take the survey, we screened individuals who were under 18 years old. The survey was only available in German, but English translations of the questions are available in the appendix.

A total of 746 individuals clicked on our link.[4] Among them, 75 completed the survey, resulting in a completion rate of 10.1%. Of the 75 completes, 24 did not provide a valid username. Depending on the analysis, we used the full 75 completed interviews or the subset of 51 completed interviews for which a username was available. This sample size limits the power of all analyses below. We advise readers to consider any analyses conducted using the Reddit Survey as preliminary and would encourage any researcher to repeat these analyses with a larger, more robust sample.

5.2.1.4 Reddit 2018 Data

For the 51 Reddit Survey respondent who provided a valid username, we used the Reddit application programming interface (API) to scrape all of their submissions made between 14 July 2017 and 13 July 2018 (the day on which the Reddit Survey closed). A total of 18 279 submissions were identified (3625 in German, 10 056 in English, and 4598 in an undetectable language). The resulting data were cleaned and coded in the same manner as the Reddit 2016 data. They were subset to English and German posts, flagged as being on or off topic, summarized by language and author, and run through the sentiment analysis models.

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4 There is no way to measure post views on Reddit, so the number of individuals who saw the link but opted not to click on it is unknown. However, Reddit's CEO, Steve Huffman, estimates that two-thirds of new Reddit users are lurkers, meaning they consume information but opt not to participate (Protalinski 2017). Many more individuals are likely active Reddit contributors but opted not to participate in our survey. For additional perspective, the median click rate for Twitter ads is 1.51% (Rodriguez 2018). We expect that this is likely comparable to the click rate for our Reddit survey.

Table 5.2 Number of entries by type and summarization metric.

	Authors with data on relevant topics			Authors with measurable sentiment – modeled			Authors with measurable sentiment – manually coded[a)]
	German	English	Total	German	English	Total	Total
Total	10	18	19	8	17	17	15
Political ideology	8	14	16	7	14	15	11
Interest in politics	9	16	17	7	16	16	14
Immigration	7	9	26	7	7	11	8
EU	3	10	12	3	10	12	3
Gay rights	3	8	9	3	8	9	3
Climate change	4	8	8	4	8	8	2

a) Hand-coded sentiment was not calculated by language, so only a total count is available.
Source: Reddit (2018).

We manually coded all 13 681 English and German submissions. Each submission was coded as a given topic, with a small subsample double-coded to evaluate coder variance.[5] We then read all items flagged as relevant for a given author and topic altogether to determine a sentiment score. Phrases such as "I voted for the AFD" resulted in sentiment codes at the ends of the spectrum, while phrases that included more than one viewpoint (e.g. "I understand why people are afraid of the effect immigrants may have on the economy, but I also want to be sympathetic to refugees in need") were assigned codes toward the middle. This coding scheme and the limited data available were prone to researcher subjectivity and cannot be considered a gold standard. This scheme can only be used to demonstrate consistency or inconsistency with the sentiment models. Table 5.2 includes the same information as Table 5.1 for the Reddit 2018 data and has additional columns for the counts of authors who could be assigned a sentiment via manual coding.

5.2.2 Analysis

For all analyses, we assumed that the Reddit data would be used in isolation (i.e. it would not be combined with survey data to create hybrid estimators). We made this assumption for three reasons. First, it is real world. As evidenced by the examples in the introduction, several researchers are attempting to use

5 We did this check at the beginning of the coding process to ensure both coders were coding the data consistently. When differences were identified, they were discussed and rules were edited/clarified before coding the rest of the entries.

social media data without supplementing with other sources. For even more examples of how Big Data (including social media) are being used independently of survey data, please see Stephens-Davidowitz (2017). Second, we believe it is important to understand all sources of error even if using social media data in combination with other data sources. Understanding the flaws in each data source independently help researchers to identify ways in which to combine data (if desired). Third, social media data could be combined with survey data in numerous ways and create more ways errors could be introduced or mitigated. We have considered several of these methods in the summary.

To address our first research objective, i.e. whether social media data can be used to produce similar attitude distributions as survey data, we compared the means, medians, and standard deviations of the sentiment scores from the Reddit 2016 data to the comparable indices from the ESS. We made comparisons using two-sample, two-sided *t*-tests on the final weighted ESS data and unweighted Reddit 2016 data.[6] Reddit is anonymous; therefore, weights cannot be constructed for any Reddit data.[7]

To answer the second question concerning the source of the observed differences, we analyzed three potential sources of error: missingness, measurement, and coding. The definitions and analysis plan for each are described in this section and summarized for easy reference in Table 5.3. Each comparison is meant to isolate, to the best of our abilities, one error source while holding all other error sources constant. Unfortunately, some of the comparisons do not entirely isolate one error source, and we note this as a limitation of the analysis and discuss the implications on our ability to draw inference in the discussion. As mentioned previously, one weakness in most of these analyses is the sample size of the Reddit Survey and the Reddit 2018 datasets. We are cognizant of this limitation and have accounted for this in our discussion of the results.

5.2.2.1 Missingness

Missingness occurs when there is no submission to be analyzed and may introduce error if the individuals who choose not to post on Reddit have different opinions than those who do post. Missingness can be further broken down into two

6 *t*-Tests are designed for random samples and may not be appropriate on nonprobability samples such as the Reddit 2016 data. We also compared the data using the Mann–Whitney *U* test, which does not rely on the same assumptions as the *t*-test but also did not allow us to apply the ESS weights (Mann and Whitney 1947; Fay and Proschan 2010). The results between the *t*-tests and Mann–Whitney tests were similar, so we opted to report the *t*-tests. We conducted a similar comparison for all other analyses mentioned in this section.

7 Although redditors are not required (or even asked) to provide personal information, algorithms have been developed to assign attributes to individuals based on their posts (see https://snoopsnoo.com). Researchers should consider the ethical implications in addition to the accuracy of these algorithms before using them.

Table 5.3 Summary of analyses

Error component	Datasets used	Test conducted
Overall	ESS vs. Reddit 2016	*t*-test
Missingness	ESS vs. Reddit Survey	*t*-test and χ^2
	Reddit Survey respondents with username vs. Reddit Survey respondents with username and topical posts	*t*-test
Measurement	Reddit Survey vs. Reddit 2018 (auto-coded sentiment scores)	Paired *t*-test and correlation
	Reddit Survey vs. Reddit 2018 (manually coded sentiment scores)	Paired *t*-test and correlation
Coding	Reddit 2018 auto-coded sentiment score vs. Reddit 2018 manually coded sentiment score	*t*-test
	Reddit 2018 auto-coded topic model "Relevance" flag vs. Reddit 2018 manually coded "Relevance" flag	κ

subcomponents. First, if individuals do not use Reddit in the first place, no text is available to analyze. For survey researchers, this type of missingness is similar to undercoverage in surveys. To assess this type of missingness, we compared the average sentiments as calculated by the Reddit Survey responses to the average sentiment scores using the ESS using two-sample, two-sided *t*-tests. We also compared demographic distributions across the two sources. Significant differences may suggest that the Reddit population is different from the ESS population, introducing bias. This comparison assumed that the Reddit Survey respondents were similar to the Reddit population as a whole, and there was no self-selection or nonresponse bias. It also assumed that any measurement error typically found in surveys (e.g. social desirability bias) was consistent in both the ESS and Reddit Survey and that the individuals who posted on Reddit would submit similarly honest responses as those who opt not to post. We discuss the likelihood of these assumptions and the effect on the overall conclusions in the results section.

Second, active redditors may not post on a given topic because they are not interested in the topic or do not want to share their opinion through this medium. In other words, they are a member of Reddit, so they have the opportunity to post but opt not to. This is similar to the concept of nonresponse in survey research. To identify whether this may be a source of error in these data, we used two-sample, two-sided *t*-tests to compare the average Reddit Survey sentiment scores among all respondents who provided a username to the average Reddit Survey sentiment scores among respondents who also submitted at least one post or comment on the topic of interest. Given that one group is a subset of the other, we used the repeated measures analysis method to account for the violation of the independence assumption (Center for Behavioral Health Statistics and Quality 2015). Significant differences would suggest that some error is attributable to this source of missingness.

5.2.2.2 Measurement

Measurement error may also be present in the Reddit 2016 data. Individuals often use social media to present the most favorable version of themselves (Van Dijck 2013). Using Reddit text data to create estimates of the strength of social attitudes will be biased if the content posted to Reddit is not representative of the individuals' views. To identify measurement error in these data, we compared the sentiment scores using the Reddit 2018 data to the survey responses using the Reddit Survey. This comparison has the potential to confound coding error with the measurement since we cannot be certain whether any observed differences in the scores are the result of misrepresentation on the part of the respondent or faulty modeling on the part of the researcher. To address this issue, we conducted the comparison using both the model-based sentiment scores and the manually coded scores. Both the model-based and manually coded sentiment scores likely contain some coding error, but if we find differences between the survey values and the values derived from text, then we can be relatively confident that at least some of the difference is caused by the redditor's misrepresentation. Comparisons were limited to individuals for which a survey value and a sentiment score were available. We conducted two types of analyses. First, we compared the average sentiment scores for all individuals across the two sources using a paired sample t-test. Second, we created a correlation coefficient to determine whether the two scores for a given individual were correlated. A significant t-statistic or low correlation coefficient would suggest measurement error.

5.2.2.3 Coding

Unlike the ESS and Reddit Survey data, the Reddit 2016 and Reddit 2018 submissions (i.e. text) had to be coded by topic and onto a scale before it could be analyzed. In theory, individuals could represent their attitudes accurately and completely online, but our attitude distributions derived from these submissions could still be biased due to poor models. The models could fail in two ways: (i) they may not appropriately identify relevant submissions or (ii) they may not correctly assign attitude strength. To assess the extent of bias introduced by coding error, we conducted two comparisons. Using the Reddit 2018 data, we first conducted a two-sided, two-sample t-test to compare the average manually coded sentiment scores to the average dictionary-coded sentiment. Although both coding methods likely have error, we assumed the error was primarily due to poor models and attributed the differences to bias in the models. Second, we further assessed the source of the bias by comparing the dictionary-based topic model flags (i.e. on or off topic) to the manually coded topic flags using a κ statistic. A high κ would indicate that the models worked well, whereas a low κ would demonstrate that the dictionary search was not identifying the correct submissions for analysis.[8]

5.3 Results

5.3.1 Overall Comparisons

The first two columns of each cluster in Figure 5.1 display the average attitude for the ESS and Reddit 2016, respectively. For most variables, the higher the value, the more support (e.g. a 10 would be complete support of the EU, while a 0 would be no support). In the case of political ideology, higher scores represent right-leaning

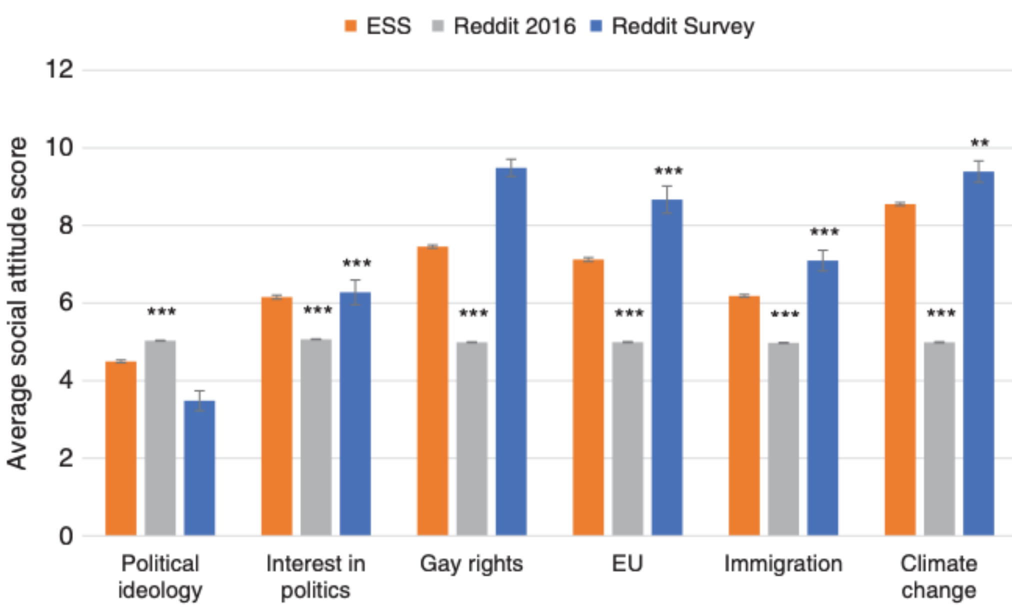

Figure 5.1 Average social attitude score. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (compared to the ESS). Source: ESS, Reddit 2016, Reddit Survey.

8 We intended to conduct a third comparison in which we subset the data to submissions that were flagged by both topic assignment methods as being on topic and calculate the correlation between the sentiment score as coded by the researchers and the sentiment score as coded using the lexicons. A high correlation would have indicated that the lexicons well represented the sentiment in the text, whereas low correlations indicated poor representation. Unfortunately, small sample sizes precluded this analysis.

views, and for climate change, a higher score means a stronger belief in its existence. There are two interesting findings from the ESS and Reddit 2016 comparison. First, the Reddit 2016 data produced significantly different averages compared to the ESS for all attitude measures. The Reddit 2016 averages were consistently more conservative than their ESS counterparts – further to the political right, less support for gay rights, less support for the EU, less favorable perceptions of immigrants and immigration, and less likely to believe in climate change. Redditors were also less interested in politics. This finding was similar when we reviewed the medians and used base-weighted ESS data (not shown). One could argue that the significance level is an artifact of the sample sizes (n ranged from 2829 to 12 826 for each group), but the magnitude of the differences is quite large in some instances (e.g. 3.56 for climate change) and would remain significant with smaller samples.

Second, there is little variability across the Reddit 2016 averages. All hover around 5.0 (the midpoint on the 11-point scale). We hypothesized that this finding may be the result of Reddit culture. As "the front page of the Internet," submissions should be about fact, not about opinion. However, we did not find support for this hypothesis. The methods used to construct the sentiment scores discarded all posts about the topic that did not contain any words in the sentiment lexicon. The only submissions included were those with some positive or negative value, so the unopinionated submissions could not have added noise to the models. Additionally, while the means did not fluctuate across variables, all variables had individuals who were coded into all possible values in the range (0–10). This suggests that people are posting opinions to Reddit, but differing opinions appear to balance out.

Based on this analysis, we were unsuccessful at achieving our first objective – to use social media data to construct a similar attitude distribution as that created from ESS data. Whereas the ESS skewed toward more liberal attitudes, the Reddit 2016 data were more centrist and normally distributed.

5.3.2 Missingness

To assess the source of the observed differences between the ESS and the Reddit 2016 data, we isolated the various error components, beginning with error caused by missingness. As mentioned in the methods section, we evaluated two types of missingness. First, we assessed error caused by individuals who do not use Reddit (i.e. the uncovered). The third column in each cluster in Figure 5.1 displays the average sentiment scores from the Reddit Survey. Overall, the Reddit Survey respondents were significantly more liberal than their ESS counterparts – they identified further to the left of the political spectrum, were more likely to believe that climate change is real, and demonstrated more support for gay rights, the EU, and acceptance of immigrants and immigration. This ideological bias is consistent

Table 5.4 Weighted percentages, means, and distributions of sociodemographic variables.

	Reddit Survey	ESS	*p*-value
Female (%)	5.0	51.0	<0.0001
Age (mean)	27.4	48.7	<0.0001
Married (%)	17.0	53.5	<0.0001
Years of education (mean)	16.6	13.6	<0.0001
Urbanicity (% distribution)			
Big city	40.0	14.4	<0.0001
Suburbs	16.7	13.8	
Town or small city	35.0	35.3	
Country village, farm, or countryside	8.3	36.5	

Source: Reddit Survey, ESS.

with the demographic distribution of the Reddit Survey respondents (Table 5.4). These respondents were significantly more likely to be male, be younger, have more education, and are more likely to live in urban areas than the ESS respondents, all attributes correlated with more liberal attitudes (Gallego 2007).

As previously noted, this analysis assumes that the Reddit Survey respondents are representative of the Reddit population. Although exact statistics are not available on the Reddit population, the distributions in Table 5.3 are consistent with Reddit's estimates of its users' sociodemographics (https://www.digitaltrends.com/social-media/reddit-ads-promoted-posts) and others' research on the topic (Shatz 2017; Singer et al. 2014; Duggan and Smith 2013). This information lends support to our assumption that the differences observed between the Reddit Survey and ESS are the result of coverage, not self-selection bias. However, our sample sizes are small and these data should not be considered conclusive.

Second, we analyzed whether there were missing attitudes among redditors. That is, are redditors who submit relevant content to Reddit different from redditors who do not submit content and are thus missing from our sentiment models? Figure 5.2 displays the average survey responses among individuals who posted on each topic and those who did not. Although none of the comparisons reached significance, this is not surprising given the small sample size. However, the magnitude of the difference among four of the seven comparisons (political ideology, gay rights, EU, and immigration) was smaller than 0.2 and varied in the direction of difference.

Given these analyses, we suspect that missingness error is large and is driven by the difference between the Reddit population and the general population

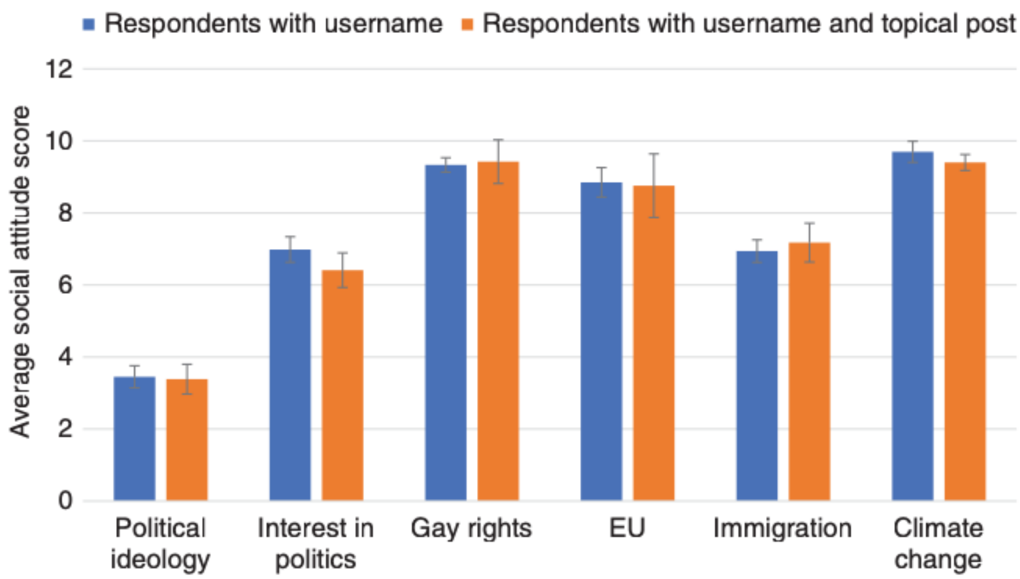

Figure 5.2 Average social attitude score among Reddit Survey respondents with a valid username by presence of submissions. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Source: Reddit Survey.

(i.e. coverage bias, in TSE). These data and others' suggest that Reddit underrepresents older, married, and lesser educated individuals, females, and those living in rural areas. As a result, Reddit posts express more liberal attitudes than the general population. The data also suggest that while the Reddit population is not representative of the general population, redditors who post about a given topic appeared to be similar to redditors who do not post. If demographics are strongly correlated with the measures of interest, then this finding suggests there will be minimal nonresponse bias. However, we are cautious about placing too much credence in this conclusion given the small sample size.

5.3.3 Measurement

Next, we attempted to isolate the contribution of measurement error to the differences between the ESS and social media attitude distributions. Figure 5.3 displays the results of our analysis – the average difference between an individual's survey response on the Reddit Survey to the same individual's sentiment value (both auto-coded and manually coded). All of the comparisons between the auto-coded sentiment scores and the survey responses were significantly different, whereas two of the five manually coded versus survey comparisons reached significance. Despite a difference in significance, the direction of the difference was consistent in five of the six variables (exception: interest in politics). These findings suggest that while we are likely conflating some coding error with measurement error, individuals are not posting material that adequately depicts their full set of

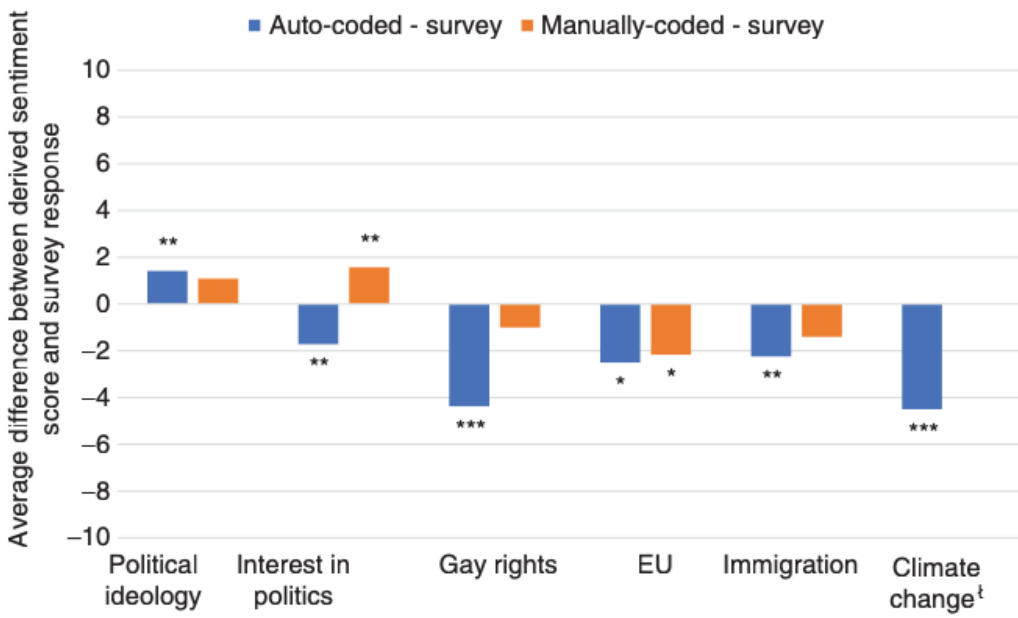

Figure 5.3 Average difference between derived sentiment score and Reddit Survey response. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. (ŧ A paired t-test was not available given the lack of variance for the manually coded values.) Source: Reddit 2018 vs. Reddit Survey.

attitudes. Redditors appear to be more liberal than their posted content would indicate. When compared to their survey responses, individuals' Reddit content skewed toward the political right, was less supportive of gay rights, the EU, and immigrants or immigration, and was less likely to agree that climate change is real. Their exclusion of their full set of views resulted in significant measurement error that contributed to the overall differences. These findings were affirmed using a correlation matrix (not shown).

5.3.4 Coding

Figure 5.4 displays the average sentiment scores produced using the sentiment models versus those produced through manual coding. Similar to the results in Figure 5.1, the auto-coded scores were centered near the midpoint of the scale (e.g. near 5) and were normally distributed. This was inconsistent with the manually coded sentiment scores that tended to show more liberal attitudes. Despite the large differences observed in Figure 5.4, only two of the six comparisons reached statistical significance – interest in politics and support for immigrants and immigration. This is a result of the exceptionally small sample sizes (n ranged from 2 to 17) and all results in this section should be interpreted with extreme caution.

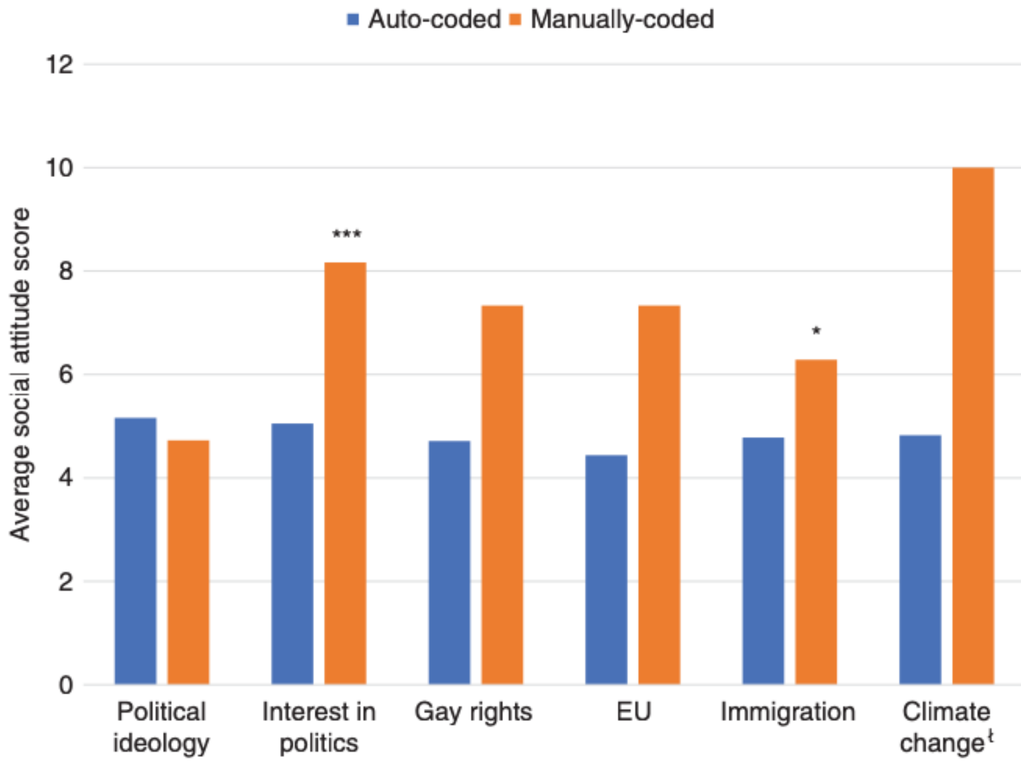

Figure 5.4 Average social attitude score by coding scheme. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. (ł A paired t-test was not available given the lack of variance for the manually coded values.) Source: Reddit (2018).

The data in Figure 5.4 only compare the averages by coding method and the net effect of coding. However, there were two steps in the coding process – (i) topic modeling in which Reddit posts were flagged as relevant or not and (ii) sentiment models applied to the relevant posts. κ statistics (not shown) were calculated to identify whether the difference in coding outcomes were a function of flagging different entries. κ values ranged from 0.096 for support of the EU to 0.515 for support of gay rights, suggesting significant disagreement between coding methods in the identification of relevant posts.

Based on these analyses, we hesitantly (due to small sample sizes) concluded that coding error further contributed to our inability to use social media data to create similar attitude distributions as those produced by the ESS. Not only were the sentiment scores different between coding methods, but these differences were the result of flagging different cases for consideration in determining sentiment. Although our ability to draw strong conclusions is limited by the data, additional qualitative evidence suggests that coding error could be sizeable. In Appendix 5.C, we walk through an example of how the topic and sentiment models were applied. We provide a sample post (for illustrative purposes only):

Post 1 : I think climate change is real.(Author 1)
    Comment 1.1 : I disagree.(Author 2)
    Comment 1.2 : I can prove it.(Author 1)

In this example, Author 1 has clearly stated that she believes in climate change. However, the AFINN lexicon used in sentiment coding would have coded her as having expressed no sentiment because none of the words used (I think climate change is real. I can prove it.) are in the lexicon. If this example is indicative of real content, the current modeling approach will suffer from significant coding error.

5.4 Summary

Unfortunately, we were unable to meet our first objective – to use social media data to create similar attitude distributions as those observed on the ESS. The Reddit data, as analyzed in this chapter, produced more centrist and normally distributed scores than the ESS, skewing estimates toward the conservative end of the spectrum on all attitude measures. This conservative skew was driven by two types of error: measurement and coding. Individuals posted more conservative sentiment on Reddit than they reported in surveys. Our models also contributed to error in that they did not identify the correct Reddit submissions for inclusion in the sentiment analysis. These findings were moderated by missingness due to undercoverage. The Reddit population was younger, more urban, and had a higher level of education than the general population (16 and over). These individuals pushed the overall sentiment scores back toward the liberal side of the distribution. However, the error introduced by missingness did not completely offset the other two errors, and the overall estimates still showed more conservative attitudes than those produced by the ESS.

Despite our best efforts to conduct thorough analyses and isolate various error components, these analyses suffer several limitations. First, and most importantly, several of these analyses have extremely small sample sizes, reducing power, and our ability to draw conclusions. Our overall analyses of the attitude distributions were conducted using sufficient sample sizes, but several of the other comparisons were conducted on as few as eight cases. Some of the analyses conducted using small case counts were further supported with anecdotal evidence or findings from other surveys, but we still caution readers from placing too much credibility in the statistics presented here. Instead, we encourage researchers to reproduce this research on a larger scale with Reddit and, in general, to be more cognizant of the potential for and sources of error in Big Data.

Second, we have insinuated that the ESS is the more accurate data source, but that assumption may be incorrect. Surveys are prone to social desirability bias, resulting in individuals reporting more liberal beliefs than they actually hold (DeMaio 1984; Goffman 1959). As a result, it is possible that the Reddit data may be less prone to some types of error than the ESS. For example, Reddit is anonymous, and individuals may feel freer in expressing less socially desirable attitudes than when sitting across from an interviewer. Although this scenario is possible, there are also limits to the freedom that anonymity provides. Germany recently passed a law that requires social media platforms, including Reddit, to remove all hate speech within 24 hours or face a fine (the Netzwerkdurchsetzungsgesetz [NetzDG]). This law could influence the types of posts or comments submitted or those available for analysis. Even if redditors felt freer to express their true perspectives, redditors are not representative of the general populations, and missingness error (i.e. coverage bias) in the Reddit data is likely to swamp any improvement in measurement error. Overall, we believe that our initial assumption is appropriate – the ESS data are more accurate than the social mediate data. However, we note this limitation because researchers should consider whether this assumption holds for their own data sources.

Third, in an ideal setting, we would have been able to further isolate the error sources. We could have done this if ESS collected information on whether respondents had a Reddit account and collected their username. We could have linked their posts to their ESS survey responses. Unfortunately, we could not, and we had to isolate the various error sources by using a combination of different data sources. Not only did this approach introduce a level of complexity into the analysis, but it also required assumptions that the various data sources suffer from similar errors. Throughout this chapter, we note where these assumptions likely hold and where they are more questionable. Researchers will continue to need to make decisions using imperfect data, and we believe that these data are a good starting point. But we continue to encourage researchers to continue to research this topic area to validate or refute these findings.

Finally, one error source we were not capable of isolating and investigating was related to geography. Data are available on the ESS to identify and limit analyses to individuals living in Germany. Given the anonymity of Reddit, we had to focus on using data from subreddits based in German-speaking countries. To the extent that we included Austrians or individuals from other countries, we have introduced error that could add noise, increase or reduce the observed differences.

Given our findings, these limitations, and the growing need for a large amount of high-quality but cheap data, we recognize researchers' desire to find ways to use social media data to replace or supplement survey attitude data. Our results suggest that Reddit may not currently be a strong candidate to create similar attitude scales as those researchers typically use on surveys such as ESS, but we encourage them to keep working on finding ways in which social media data can be used for attitudinal research. First, researchers should focus on improving the models and modeling methods. We used a simple lexicon to conduct our analyses since this is a common approach by social scientists. However, there are multiple types of topic and sentiment models, each with strengths and weaknesses (Choi and Lee 2017). More research is necessary to evaluate and improve these models so they are appropriate for short entries (e.g. tweets), accommodate the social media lexicon, and account for sarcasm, emojis, and punctuation. Many of these models are also limited to processing one language at a time. To the extent to which language is nontransferrable – that equivalent words are not used to express equivalent attitudes – sentiment models that accommodate multiple language will be subject to error. These models were also limited to posts. Researchers may investigate methods and the utility of incorporating additional data such as up/down votes, likes, and retweets to improve the accuracy of these models. As these sentiment and topic models improve, we expect that the ability to use social media in attitudinal research will also improve.

Second, researchers should continue to consider and test for various error components when assessing the feasibility of using Big Data as a supplement or replacement for survey data. Even when overall estimates produced from Big Data appear consistent with survey estimates, competing, and canceling error sources may exist. We must understand these errors to identify how they may affect the estimate's variance or alter the correlations between variables. These types of analyses also increase the usefulness to other researchers who may have data with slightly different attributes. By breaking down error sources, they may be able to identify the reason their results may differ and, ideally, improve their methods.

Third, and counter-intuitively, we encourage researchers to wait patiently. As evidenced by this chapter, a significant amount of the differences between Reddit and the ESS attitude scales were the result of missingness from undercoverage. The effect of undercoverage will change over time as some social media platforms become more or less popular and as more individuals gain access to the Internet. Reddit is the fifth most popular website in Germany as of October 2018, but we estimate that only 12.9% of the population uses it (Alexa 2018),[9] Just as telephone penetration had to reach a critical mass before telephone surveys were able to create similar estimates as their face-to-face counterparts, social media needs time to gain further popularity and uptake.

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9 Exact usage statistics are not available. We calculated 12.9% using the following formula: number of active Reddit users per month (330 million) * proportion of Reddit page views that originate from Germany (3.24%)/German population (82.79 million; https://www.redditinc.com, https://www.similarweb.com/website/reddit.com). This estimate should be used only to illustrate the vast growth potential for social media.

Fourth, researchers may accept the error and account for it by developing or applying different analytic methods. For example, survey researchers typically use frequentist models when imputing missing values. These models assume limited error in the observed data on which they are based. However, we know that this assumption does not hold on the Reddit data. Researchers are experimenting with the application of Bayesian models to avoid the limited error assumption, more accurately impute missing values, and create more accurate overall estimates (Arima, Datta, and Liseo 2015). This work has been limited to other applications to date (small area estimation is discussed in the above cite), but it could be applicable to Big Data as well. This research is a work in progress and only one example of a method that accepts the error and adjusts other steps in the estimation process to accommodate and correct for it. These methods are encouraging, but they reinforce the need to understand the type of error that exists in the social media (or any Big Data source). Without understanding the type of error, researchers will not know which model assumptions may be violated and which types of models may be adapted to correct for the error.

Finally, we encourage researchers to look into alternative uses of social media data for which they may be better suited. For example, hybrid estimates that are created from a combination of social media and surveys may allow researchers to reduce costs over current designs that exclusively use survey data but minimize the risk of bias found in social media data. Even though it is not an example of using social media data, Oberski and his colleagues create hybrid estimates from administrative and survey data using generalized multitrait-multimethod models. This approach accounted for both random and systematic errors and could accommodate the types of distributions that occur in nonsurvey records (e.g. nonlinearity; Oberski et al. 2017). Alternatively, social media may be used to enhance survey efficiencies. Topic models could be used to identify rare populations (e.g. smokers, individuals with a rare disease). Once identified, individuals may opt to reach out to the identified individuals and invite them to participate in a survey. Or they could separate all individuals in the topic models into two strata – those likely and unlikely to be in the target population – and use this to draw a stratified sample. If used to supplement frame construction or to stratify a sample, the error we observed in the models may be acceptable or could be accounted for in the survey weights. These and other methods to use Big Data and survey data in tandem are discussed in more detail by Lohr and Raghunathan (2017). Given the anonymity of Reddit (no information is required to open a Reddit account), weighting was not possible while using Reddit in isolation. We hope that researchers do not finish this chapter deflated but with a greater respect for the potential errors in Big Data and the goal of understanding those errors when choosing a data source or analysis plan.

5.A 2016 German ESS Questions Used in Analysis

	German wording	English wording	Number of scale points
Political interest	Wie sehr sind Sie an Politik interessiert?	How interested are you in politics?	4
Political ideology	In der Politik spricht man manchmal von “links” und “rechts.” Wo auf der Skala würden Sie sich selbst einstufen, wenn 0 für links steht und 10 für rechts?	In politics people sometimes talk of “left” and “right.” Where would you place yourself on this scale, where 0 means the left and 10 means the right?	11
Gay rights	Bitte sagen Sie mir, wie sehr Sie jeder der folgenden Aussagen zustimmen oder wie sehr Sie diese ablehnen: Schwule und Lesben sollten ihr Leben so führen dürfen, wie sie es wollen.	Please say to what extent you agree or disagree with each of the following statements: Gay men and lesbians should be free to live their own life as they wish.	5
	Bitte sagen Sie mir, wie sehr Sie jeder der folgenden Aussagen zustimmen oder wie sehr Sie diese ablehnen: Wenn ein nahes Familienmitglied schwul oder lesbisch wäre, würde ich mich schämen.	Please say to what extent you agree or disagree with each of the following statements: If a close family member was a gay man or a lesbian, I would feel ashamed.	5
	Bitte sagen Sie mir, wie sehr Sie jeder der folgenden Aussagen zustimmen oder wie sehr Sie diese ablehnen: Schwule und lesbische Paare sollten die gleichen Rechte haben, Kinder zu adoptieren, wie Paare, die aus Mann und Frau bestehen.	Please say to what extent you agree or disagree with each of the following statements: Gay male and lesbian couples should have the same rights to adopt children as straight couples.	5
EU	Jetzt kommen wir zum Thema Europäische Union. Manche Leute sagen, dass die europäische Einigung weiter gehen sollte. Andere sagen, dass sie schon jetzt zu weit gegangen ist.	Now thinking about the European Union, some say European unification should go further. Others say it has already gone too far.	11
	Stellen Sie sich vor, morgen würde eine Volksabstimmung in Deutschland über die Mitgliedschaft in der Europäischen Union stattfinden. Würden Sie für die Fortsetzung der Mitgliedschaft Deutschlands in der Europäischen Union oder für einen Austritt Deutschlands aus der Europäischen Union stimmen?	Imagine there were a referendum in Germany tomorrow about membership of the European Union. Would you vote for Germany to remain a member of the European Union or to leave the European Union?	2

	German wording	English wording	Number of scale points
Immigration	Nun geht es zunächst um die Zuwanderer, die derselben Volksgruppe oder ethnischen Gruppe angehören wie die Mehrheit der Deutschen. Wie vielen von ihnen sollte Deutschland erlauben, hier zu leben?	Now it is first about the immigrants which have are of the same race or ethnic group as most Germans. To what extent do you think Germany should allow them to come and live here?	4
	Wie ist das mit Zuwanderern, die einer anderen Volksgruppe oder ethnischen Gruppe angehören als die Mehrheit der Deutschen?	How about people of a different race or ethnic group from most German people?	4
	Und wie ist das mit Zuwanderern, die aus den ärmeren Ländern außerhalb Europas kommen?	How about people from the poorer countries outside Europe?	4
	Was würden Sie sagen, ist es im Allgemeinen gut oder schlecht für die deutsche Wirtschaft, dass Zuwanderer hierher kommen?	Would you say it is generally bad or good for Germany's economy that people come to live here from other countries?	11
	Würden Sie sagen, dass das kulturelle Leben in Deutschland im Allgemeinen durch Zuwanderer untergraben oder bereichert wird?	Would you say that Germany's cultural life is generally undermined or enriched by people coming to live here from other countries?	11
	Wird Deutschland durch Zuwanderer zu einem schlechteren oder besseren Ort zum Leben?	Is Germany made a worse or a better place to live by people coming to live here from other countries?	11
Climate change	Sie haben vielleicht von der Auffassung gehört, dass sich das Klima auf der Erde verändert, weil die Temperaturen über die letzten 100 Jahre gestiegen sind. Wie ist Ihre persönliche Meinung dazu? Denken Sie, dass sich das globale Klima gegenwärtig verändert?	You may have heard the idea that the world's climate is changing due to increases in temperature over the past 100 years. What is your personal opinion on this? Do you think the world's climate is changing?	4
	Wie viel haben Sie vor dieser Umfrage über den Klimawandel nachgedacht?	How much have you thought about climate change before this survey?	5

5.B Search Terms Used to Identify Topics in Reddit Posts (2016 and 2018)

5.B.1 Political Ideology

polid = "(links & politik) | (left & politic) | (rechts & politik) | (right & politic) | extremismus | nazis | (links & extrem) | (left & extrem) | (rechts & extrem) | (right & extrem) | antifa | faschist | rechter flügel | right.wing | rightwing | afd | nationalismus | nationalism | heimatliebe | homeland love | linker flügel | left.wing | leftwing | nsu | rechtsruck | right jerk | linksruck | left jerk | konservativ | conservatism | spd | die grüne | green party | sozialdemokratie | social democracy | csu | cdu | linkspartei | mitte.links | middle.left | demokrat | democrat | fdp | liberal | sozialist | socialist"

5.B.2 Interest in Politics

polint = "(links & politik) | (left & politic) | (rechts & politik) | (right & politic) | extremismus | nazis | (links & extrem) | (left & extrem) | (rechts & extrem) | (right & extrem) | antifa | faschist | rechter flügel | right.wing | rightwing | afd | nationalismus | nationalism | heimatliebe | homeland love | linker flügel | left.wing | leftwing | nsu | rechtsruck | right jerk | linksruck | left jerk | konservativ | conservatism | spd | die grüne | green party | sozialdemokratie | social democracy | csu | cdu | linkspartei | mitte.links | middle.left | demokrat | democrat | fdp | liberal | sozialist | socialist | erstwähler | geht wählen | go vot | wahlbeteiligung | vote | politic | demokratie | democracy | verfolgen aktueller nachrichten | (follow & news) | teilnahme an wahl | participating in election | parteimitgliedschaft | party member | beteiligung an demonstrationen | participation in demonstration | merkel | schulz | nahles | seehofer | steinmeier | maas | von der leyen | scholz | altmeier | gabriel | schäuble | de maiziere | kretschmann | gauland | weidel | lucke | helmut kohl | brandt | schröder | adenauer"

5.B.3 Gay Rights

gay = "ehe für alle | marriage for everyone | schwul | gay | lesbisch | lesbian | homosexualität | homo | lieb doch wen du willst | love whoever you want |

csd | christopher street day | bunt | colorful | regenbogen fahne | rainbow flag | gleichgeschlechtlich | same.sex | tunte | pansy | schwuchtel | faggot | kampflesbe | combat lesbian | dyke | frühsexualisierung | early sexual | sexuelle orientierung | sexual orientation | sexuelle vorlieben | sexual preference | bisexu | queer | analverkehr | sodomy"

5.B.4 EU

EU = "european uni | europäische einigung | pulse of europe | pulse of europe | zukunft europas | future of europe | pro.europ | junge europäische förderalisten | young european federalits | jef | europäische uni | internationale solidarität | international solidarity | brexit | eu.austritt | eu.withdrawal | nato | schengen | außengrenzen | external borders | blaue karte | blue card | abendland | occident"

5.B.5 Immigration

immig = "balkan route | refugee | immigra | foreigner | national community | we are the people | pegida | dominant culture | asylum | deportation | migration | immigrat | nigger | subhuman | moslems | men with kni(v|f)es | guests of merkel | people exchange | girl with headscarf | islamization | immigrant | we make it | balkanroute | obergrenze | flüchtling | zuwanderung | ausländer raus | volksgemeinschaft | wir sind das volk | leitkultur | asyl | abschiebung | asylkrisen | eger | untermenschen | muselmanen | musels | kanacken | messermänner | merkels gäste | volksaustausch | kopftuchmädchen | islamisierung | immigrant | einwander | wir schaffen das"

5.B.6 Climate

clmt = "erderwärmung | global warming | temperaturanstieg | temperature increase | klima | climate | energiewende | energy transition | eisbären | icebear | (pole & schmelzen) | (polar & melting) | co2 | kohlenstoffdioxid | carbon dioxide | treibhausgase | greenhouse gases | ozonschicht | ozone layer | umweltzerstörung | environmental destruction | dieselskandal | diesel scandal | ökoterroristen | ecological terrorists | umweltschütz | environment | naturschutz | nature protection"

5.C Example of Coding Steps Used to Identify Topics and Assign Sentiment in Reddit Submissions (2016 and 2018)

Original Post and Associated Comments

Post 1 : I think climate change is real.(Author 1)

Comment 1.1 : I disagree.(Author 2)

Comment 1.2 : I can prove it.(Author 1)

Post 2 : I like cats.(Author 3)

Comment 2.1 : Me too!(Author 4)

Collapsed Post and Associated Comments to Identify Topic, Remove Punctuation, and Convert to Lowercase

Entry 1: i think **climate change** is real i disagree
Entry 2: i like cats me too

* "climate change" is a keyword, so the algorithm would label Post 1 and all associated comments as being about climate change. There are no key words in Entry 2, so Post 2 and all associated comments would be discarded from additional analysis.

Subset Relevant Posts and Associated Comments and Transpose by Author

Author 1: i think climate change is real i can prove it
Author 2: i disagree

Tokenize Entries for Sentiment Analysis

Author 1: i, think, climate, change, is, real, i, can, prove, it
Author 2: i, disagree

* commas delineate tokens (they should not be considered punctuation).

Apply AFINN Lexicon to Obtain Initial Sentiment Score

Author 1: i, think, climate, change, is, real, i, can, prove, it = N/A (no words with sentiment score in lexicon)
Author 2: i, disagree = −2 ("disagree" has a value of −2)

Shift Scale from −5 to 5 to 0–10 for Final Sentiment Score

Author 1: N/A
Author 2: $-2 + 5 = 3$

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